AI and Economics.

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Department of Economics, The Ohio State University
Lunch & Learn with Economics Faculty, 30 October 2026



Ten weeks of my work, 20 August to 29 October

- 286 agent sessions on three machines; 3,476 instructions typed by me
- 65,700 actions by the agents: commands run, files edited, searches made
- Agents wrote about 42 million words (~530 novels) and read about 12 billion



Claude Code and Codex logs on my three machines, 29 Oct 2026; instructions are messages I typed; weekly bars 20 Aug to 28 Oct.

Since February, I have not written a line of code.

I still choose the questions, and I still decide whether I believe the answers, and what to do with them.

Is this what an economist is for?

Today

1. What AI can do, state of the art as of October 2026
2. What it is doing to the economy
3. The push for AI for science
4. What it means for economics research
5. What it means for you
6. A new undergrad econ course: ECON 5910

1. What AI can do, October 2026

In 2023 you asked a chatbot a question.

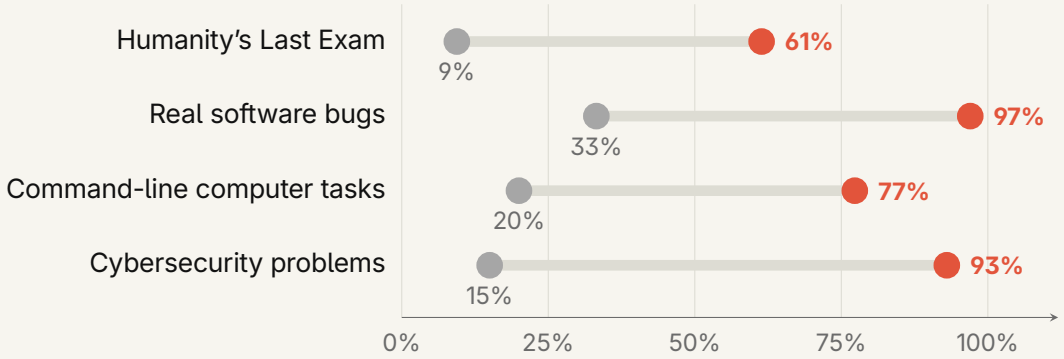
In 2026 you give an agent a job.

The exams are falling

- Math Olympiad, July 2025: a gold-medal score, graded by the official judges
- Humanity’s Last Exam, 2,500 expert questions written to stump AI: best model 9% in January 2025, 61% this month
- MIT, August 2026: AI can credibly complete almost any undergraduate written assignment

IMO 2025 (Google DeepMind); HLE paper (Jan 2025) and Artificial Analysis (Oct 2026); MIT Ad Hoc Committee report (Aug 2026).

Two years, four benchmarks



Gray: first score, 2024 to early 2025. Red: latest score, 2026.

Humanity's Last Exam: Phan et al. (Jan 2025), Artificial Analysis (Oct 2026). SWE-bench Verified: OpenAI (Aug 2024), Vals.ai (Sep 2026). Terminal-Bench and cybersecurity: Stanford AI Index 2026.

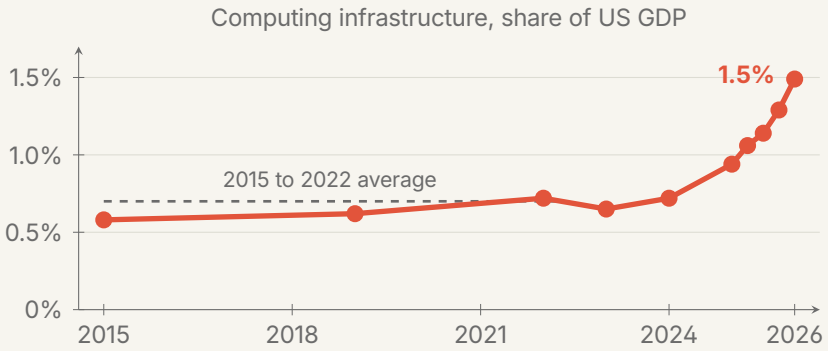
Now it does the work

- Professional tasks from 44 occupations, graded blind against experts' work: the best model wins or ties 85% (April 2026)
- Real software bugs: 97% fixed; the standard test is now retired as too easy
- Tasks that take a skilled person two working days: agents finish about half, and the benchmark cannot measure beyond that

GDPval (OpenAI, Apr 2026); SWE-bench Verified (Vals.ai, Sep 2026); METR Frontier Risk Report (May 2026).

2. What it is doing to the economy

The investment boom



About a third of US growth this year is AI-related investment, net of imports.

Epoch AI (Jun 2026; computers above trend, data center construction, networking; Q1 values and 2025 quarters). ING, Knightley (Aug 2026; Q2 2026, year over year).

Adoption is wide and shallow

- 62% of US adults use generative AI; 45% of workers use it at work
- Only 40% of occupations, and under 3% of job tasks, have a majority of workers using it
- 18 to 20% of firms use AI in a business function

Bick, Blandin, Deming, Schumacher (NBER, Aug 2026); Census Bureau BTOS (May 2026).

Jobs and pay: what the evidence says

- No detectable effect on aggregate jobs or pay yet
- Slower entry-level hiring in exposed occupations; the cause is contested
- Productivity gains in experiments; who keeps them depends on expertise

Humlum and Vestergaard (2026); Fairlie and Wu (2026); Brynjolfsson, Chandar, Chen (2026); NY Fed (2026); Brynjolfsson, Li, Raymond (QJE 2025); Autor et al. (2026).

Your cohort

- Payroll data: workers aged 22 to 25 in the most AI-exposed jobs are 19% below the path of their less-exposed peers
- Survey data: recent-graduate unemployment was 7.3% this summer, in line with 2022 to 2025, and not significantly linked to AI exposure
- Read together: exposed jobs hire fewer young workers, but graduates are not yet more often unemployed

Brynjolfsson, Chandar, Chen, “Canaries in the Coal Mine?” (Aug 2026; ADP payroll through June 2026); Fairlie and Wu, NBER w35796 (Sep 2026; CPS).

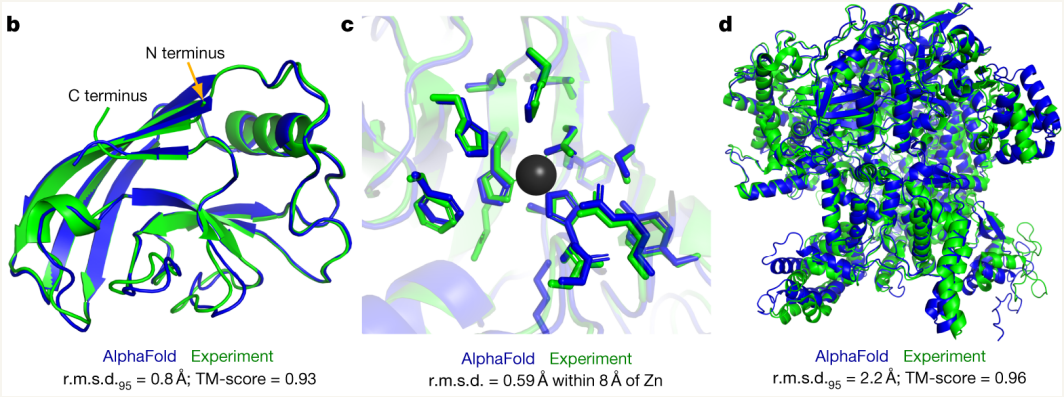
3. The push for AI for science

AI for science: the push

- Government: the Genesis Mission, over $5 billion across 278 federal projects, to “double America’s R&D output”
- Industry: $2.4 billion more pledged this month, mostly computing credits; Google, OpenAI, and Anthropic each now run a science program
- The model they all point to: AlphaFold, a Nobel Prize in 2024

White House (22 Jul and 8 Oct 2026); Google, OpenAI, Anthropic announcements (2025 to 2026); Nobel Prize in Chemistry 2024.

AlphaFold's prediction against the lab's measurement



Predicted structure in blue, experimentally determined structure in green. Jumper et al., *Nature* 596 (2021), Fig. 1b–d; CC BY 4.0, via Wikimedia Commons.

AI for science: results, and disputes

- Biology: AI-proposed drug candidates in *Nature*; not yet tested in the lab
- Mathematics: 18 open Erdős problems solved by AI; 10 claims were wrong
- Navier-Stokes, a million-dollar problem: an AI claim, still disputed
- (The most-cited 2024 study of AI and scientific productivity was withdrawn)

Nature (19 May 2026); Tao's Erdős wiki (to 30 Jun 2026); Quanta and NPR (Sep 2026); MIT statement (May 2025).

In mathematics, the check is immediate and certain:
a machine can verify a proof, step by step.

Lean, a proof assistant, accepts a proof only if every step is correct.

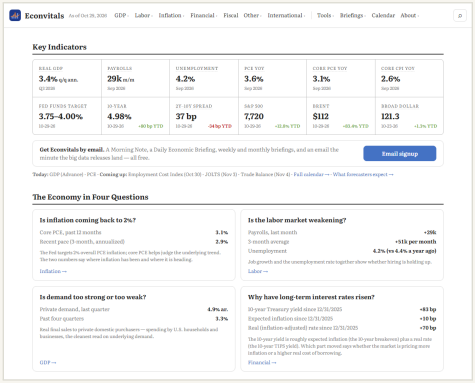
4. What it means for economics research

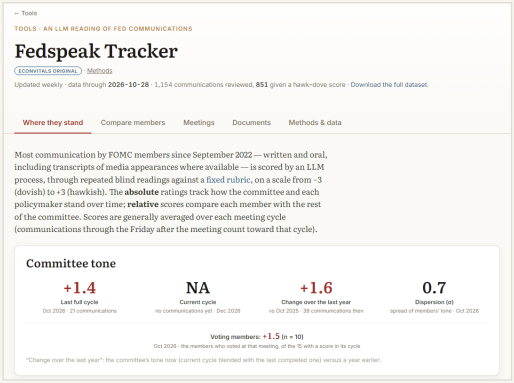
Economists are early adopters

- Using an agent weekly: economists 39%, political scientists 25%
- Students and postdocs: more than twice the rate of tenured faculty
- Agent users report about 75% more working papers posted

Anthropic survey of 1,260 quantitative social scientists (May 2026; not representative).

econvitals.org





Screenshots of econvitals.org and its Fedspeak tracker, 29 October 2026.

Econvitals, launched 23 October

Jason Furman, Harvard, formerly Obama's chief economic adviser:

- A free site tracking the US economy: 110 data sources, 790 charts
- About 484,000 lines of code, written by AI "without my direct edits"
- Briefings: written by AI, checked automatically, sent without a human read

Econvitals, as quoted by DeLong (27 Oct 2026); TokenPost (26 Oct 2026).

The discussion

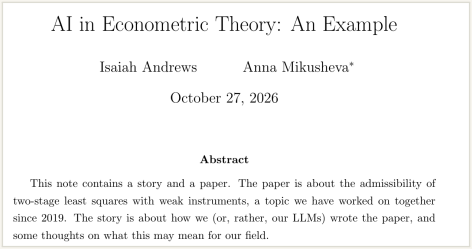
- Furman: for much of the Council's routine work, he "would somewhat regretfully and sadly choose the AI systems" over 20 economists
- Jared Bernstein, chair under Biden: the Council's work is "highly bespoke"
- Olivier Blanchard: it shows "the power of AI, when well domesticated"
- Brad DeLong: "These things do drift without careful attention"

Axios (23 and 26 Oct 2026), via TokenPost (26 Oct 2026); Blanchard on X (Oct 2026); DeLong, Substack (27 Oct 2026).

With AI, the senior economist does more.

The work it takes over is the work junior economists used to do.

Posted on arXiv, Monday 26 October



AI in Econometric Theory: An Example

Isaiah Andrews Anna Mikusheva*

October 27, 2026

Abstract

This note contains a story and a paper. The paper is about the admissibility of two-stage least squares with weak instruments, a topic we have worked on together since 2019. The story is about how we (or, rather, our LLMs) wrote the paper, and some thoughts on what this may mean for our field.

a combination of GPT-6 Astra and Claude Opus 5.5 to expand the results into the note below, which is entirely AI-written, though we asked for revisions in some places and take full responsibility for any errors.[3] Were we to write the note ourselves, we would make a number of further edits for clarity, emphasis, and style, but we provide it as-is in the spirit of illustrating current capabilities.

Andrews and Mikusheva, arXiv 2610.35831, pp. i–ii; highlight added.

Andrews and Mikusheva, MIT

- Since 2019: a related paper needed extra conditions; rejected several times; they could not drop them
- 2026: GPT-6 Astra dropped them. With weak instruments, two-stage least squares, the IV estimator from your econometrics class, is inadmissible
- That is, other estimators have lower error at every parameter value
- Proof checked in Lean; the authors take responsibility for any errors

Andrews and Mikusheva, "AI in Econometric Theory: An Example," arXiv 2610.35831 (26 Oct 2026).

Three or more instruments; squared error loss.

What it could not do, and their conclusion

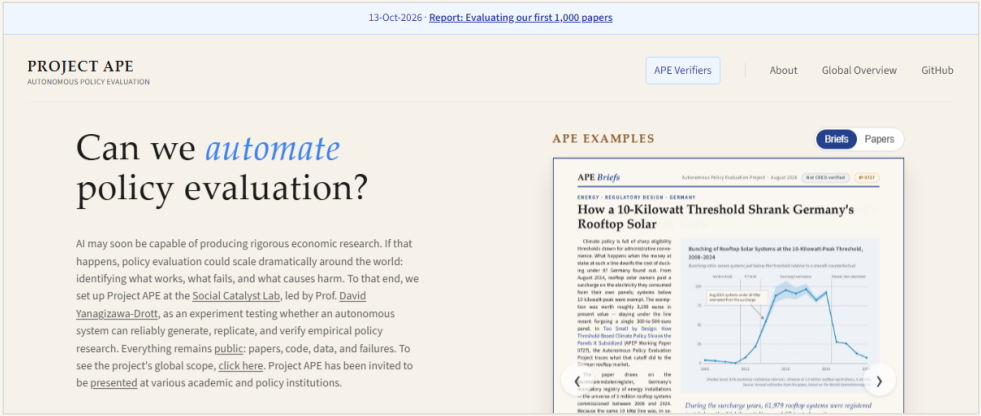
- Still open: a best estimator, one that beats it and cannot itself be beaten
- Their conclusion calls for “a serious and ongoing re-evaluation of how our field assesses research outputs” and allocates resources
- Not a tragedy, they say; but institutions have to respond
- Andrews to his PhD students, in April: model output “can be wrong in ways that require expertise to detect”

Andrews and Mikusheva (26 Oct 2026); Andrews, “Some Thoughts on AI and Research” (4 Apr 2026).

If AI gets great at coding, writing, and proofs, and stays weak at taste and judgment, the returns go to choosing the problem.

Isaiah Andrews, advice to his PhD students, April 2026.

Project APE, Zurich



Screenshot of ape.socialcatalystlab.org, 29 October 2026.

One thousand autonomous papers

- 1,000 policy-evaluation papers in four months, about $50 each
- The audit: 4.3 defects per paper; every paper flagged; 842 with a severe one
- Against published papers, humans win 88%; most of the gap is the idea

Project APE, Zurich (audit of 13 Oct 2026); Li (Apr 2026). “I don’t trust any of the papers yet, and nobody should.”

The machine checks us

- 4,452 replication packages from five journals, rerun by AI: 3,460 flagged
- Computation cut more than tenfold in 496; a new extension in 923
- Every conditionally accepted paper at the Econometric Society journals now gets an AI check; the AEA journals ran a pilot

Schwartz, Andrews, Shapiro (NBER, Sep 2026), flags, not adjudicated errors; Econometric Society editorial policies (2026); Refine (Aug 2026).

AI makes answers easy to get.

The economist chooses the question, decides whether to believe the answer, and what to do with it.

(for now…?)

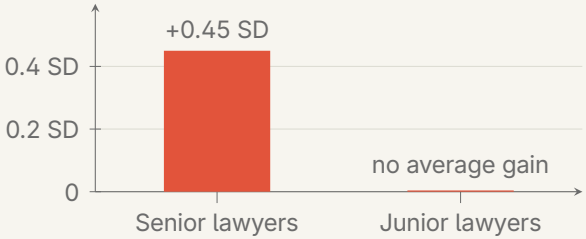
5. What this means for you

The good news:

AI raises the value of what you learn here.

AI multiplies the value of expertise

- 133 patent lawyers with an AI drafting assistant: quality rose for everyone, juniors most
- Then the assistant was removed. The lead over lawyers who never had it belonged entirely to the seniors:



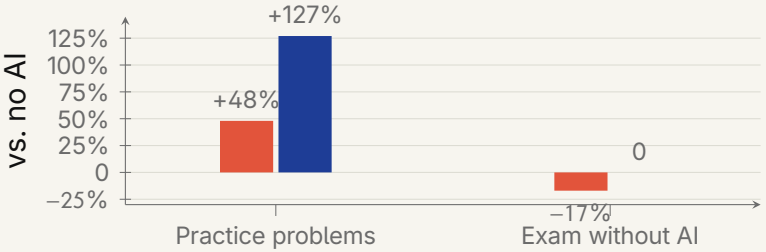
Autor, Rodchenko, Martin, Iscenko, Strand, Pearl, Ferere, NBER w35720 (Sep 2026), randomized, eleven firms, blind expert grading.

School is where expertise gets built now.

Juniors used to pay for training with routine work; agents do routine work now.

Garicano and Rayo, 2026.

AI helps you learn, if it makes you do the work



Red: plain GPT-4. Blue: GPT-4 as a tutor that only gives hints.

Harvard physics: an AI tutor that makes you produce the work beat active-learning class time.

Bastani et al., PNAS 2025, about 1,000 high-school students; Kestin et al., Scientific Reports 2025.

6. A new econ course: ECON 5910

Machine Learning & AI in Economics

- Empirical: hands-on with real economic data, the fuel of ML and AI
- Foundations: prediction, model evaluation, and how a language model works and fails, built on your econometrics
- Agents: workflows that prepare data, write code, run analysis, and report, while you set the question and judge the answer

Questions like these

- What will this house sell for?
- Which firms talk about AI in their earnings calls, and how much?
- If we raise the price, what happens to sales, and for whom?

AI agents can answer all these questions. The course teaches you the foundations for where those answers come from, and how you can start to judge them.

What you do

- Little or no programming assumed; guided Python with AI assistance
- Problem sets discussed in class; a group project defended orally
- AI use permitted and documented, with the bar raised to match
- Build on all the econometrics and economics you already know
- It is demanding: expect to work a lot to absorb the foundations, and to learn new tools and workflows

Prerequisites: introductory econometrics and statistics. Spring 2027.

You are starting economics at a good time.

Thank you. Questions, then pizza.

magnolfi.1@osu.edu

Slides: lorenzomagnolfi.com/files/ai-econ-2026.pdf

Made with Claude Code over two days: about 50 instructions from me, 1,200 agent actions, 500 web searches and page reads. Every number links to its source on the pages that follow.

Sources (1 of 3), every entry is a link

Part 1. What AI can do

IMO 2025 gold, Google DeepMind (coverage, 21 Jul 2025)
Phan et al., Humanity's Last Exam, arXiv 2501.14249 (Jan 2025)
Artificial Analysis, HLE leaderboard (Oct 2026)
MIT Ad Hoc Committee report, via the Washington Post (Aug 2026)
OpenAI, Introducing SWE-bench Verified (Aug 2024)
Vals.ai, SWE-bench Verified leaderboard (Sep 2026)
Stanford HAI, 2026 AI Index takeaways (Apr 2026)
OpenAI, GDPval
METR, Frontier Risk Report (May 2026)

Part 2. The economy

Epoch AI, computing infrastructure and GDP (Jun 2026)
ING THINK, Knightley (Aug 2026)
Bick, Blandin, Deming, Schumacher, NBER w35677 (Aug 2026)
Census Bureau, Business Trends and Outlook Survey (May 2026)
Humlum and Vestergaard, NBER w33777 (2026)
Fairlie and Wu, NBER w35796 (Sep 2026)
Brynjolfsson, Chandar, Chen, Canaries in the Coal Mine? (Aug 2026)
NY Fed, Liberty Street Economics (May 2026)
Brynjolfsson, Li, Raymond, QJE (2025)

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Part 3. AI for science

White House, Genesis Mission (Jul 2026)
White House, science summit fact sheet (Oct 2026)
Google Research at I/O (May 2026)
MIT Technology Review, OpenAI for Science (Jan 2026)
Anthropic, Claude Science (Jun 2026)
Nobel Prize in Chemistry 2024
Jumper et al., *Nature* (2021); image via Wikimedia Commons
Nature AI-scientist papers, press release (May 2026)
Tao's Erdős problems wiki
Quanta (Sep 2026)
NPR (Sep 2026)
MIT withdraws the materials paper (May 2025)

Part 4. Economics research

Anthropic, Coding agents in the social sciences (May 2026)
Econvitals
Econvitals Fedspeak tracker
DeLong, Substack (27 Oct 2026)
TokenPost (26 Oct 2026)
Axios (23 Oct 2026)
Axios (26 Oct 2026)
Blanchard on X (Oct 2026)

Sources (3 of 3)

Part 4. Economics research (continued)

Andrews and Mikusheva, arXiv 2610.35831 (Oct 2026)
Andrews, Some Thoughts on AI and Research (Apr 2026)
Project APE, Zurich
Li, arXiv 2604.03338 (Apr 2026)
Schwartz, Andrews, Shapiro, NBER w35782 (Sep 2026)
Econometric Society, editorial procedures and policies
Refine, journal partnerships (Aug 2026)

Part 5. What it means for you

Autor et al., NBER w35720 (Sep 2026)
Bastani et al., PNAS (2025)
Kestin et al., Scientific Reports (2025)
Garicano and Rayo, CEPR DP 20634 (2026)